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MANUSCRIPT

Integrating accounting and multiplicative calculus: an effective estimation of learning curve

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Abstract Numerical interpolation methods are essential for the estimation of nonlinear functions and they have a wide range of applications in economics and accounting. In this regard, the idea of using interpolation methods based on multiplicative calculus for suitable accounting problems is self-evident. The purpose of this study, therefore, is to develop a way to better estimate the learning curve, which is an exponentially decreasing function, based on multiplicative Lagrange interpolation. The results of this study show that the proposed multiplicative method of learning curve provides more accurate estimates of labour costs when compared to the conventional methods. This is because the exponential functions are linear in multiplicative calculus. Furthermore, the results reveal that using the proposed method enables cost and managerial accountants to better calculate both cost of unused capacity and product cost in a cumulative production represented by a nonlinear function. The results of this study are also expected to help researchers, practitioners, economists, business managers, and cost and managerial accountants to understand how to construct a multiplicative based learning curve to improve such decisions as pricing, profit planning, capacity management, and budgeting.

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1 Introduction

A learning curve is a mathematical function that measures how the amount of input, which is mainly direct labour hours needed to produce one unit of output, decreases as quantity of output produced increases (Nadeau et al. 2010). The learning curve, which emphasizes that individuals become more efficient at doing a task when it is performed repeatedly, is one of the most popular and generally accepted cost estimation theories (Womer and Patterson 1983; Jarkas and Horner 2011).

Identifying how costs vary in accordance with the output level is essential in a decision making process. However, since the costs act in different patterns under different conditions, predicting them is a very difficult task. Managers can use the past data to predict the costs that are expected to be incurred. An increase in efficiency of the labour force, on the other hand, may result in misleading cost predictions. This effect, in the literature, is known as the 'Learning Effect' (Drury 2012). Thus, the learning effect is a vital concept for understanding and estimating the costs in an accurate way. In this regard, with better understanding of learning effect, direct labour and production costs can be computed more accurately. This, in turn, helps managers to make business decisions, such as cost and capacity management, pricing, and human resource planning, more efficiently.

There has been an increasing interest in the application of the learning effect in accounting and industrial settings (Liao 1982; Jaber and Guiffrida 2008). The learning effect can be applied to various industries, such as aircraft, aerospace, assembly, automotive, electronics, shipbuilding, construction, defence, and petroleum refining, in which the activities are repetitious and identical (Steven 1999; Jarkas and Horner 2011).

The learning curve, which has a great practical importance, could be a useful tool for decision makers (Fioretti 2007a, b). Accordingly, there has been a stream of research investigating the learning effect in product cost evaluation, product pricing, profit planning, and control (e.g. Wright 1936; Harvey 1976; Mclntyre 1977; Liao 1982). To illustrate, Lieberman (1984) used the data on 37 chemical products and found a strong and consistent learning effect in the chemical industry. In another study, Ören and Erol (2009) analyzed the role of learning in reducing the cost of production and concluded that learning should be considered as a tool that reduces the cost of a product or service. Additionally, McKee and McKee (2013) explored the effect of the learning on revenue recognition. The authors note that the different amounts of revenue might be recognized when the learning effect is taken into account under the revenue recognition standard. Togo (2001), on the other hand, emphasized that the log-linear function should be used for learning curve because all cost behaviour is not linear. Apart from these studies, others have investigated the applicability of the learning curve in various sectors ranging from the aircraft industry to the food industry (e.g. Hartley 1965; Reis 1991; Liao 2009; Jarkas and

Horner 2011; Omah et al. 2012). The results found in those studies demonstrate that the learning curve phenomenon can be beneficially applied as a management tool.

Learning curves are key elements of strategies of some big companies such as Dell Computer, Wal-Mart, McDonald's, and Boeing. This is because these companies utilize the learning curves to decrease their costs, plan resource requirements, and increase profitability (Lanen et al. 2011; Horngren et al. 2012). According to Horngren et al. (2012), many companies such as Pizza Hut, Home Depot, and The Nissan Motor Company benefit from the learning curve in performance evaluation.

Apart from these considerations, managers use cost functions to understand cost behaviours. Cost functions are classified as linear and nonlinear cost functions (Horngren et al. 2012). One of the main reasons for a cost function to be nonlinear is the learning effect. According to the learning effect, employees are learning and becoming more familiar with their jobs which enable them to increase their efficiency. Improved efficiency, in turn, decreases unit cost. In this context, nonlinearity should be considered in the process of predicting unit costs (Horngren et al. 2012). However, it is conventionally assumed that the learning rate used in the learning curve is constant (Blocher et al. 2010). Therefore, it can be concluded that conventional learning curve model may not meet the needs of managers especially when the data have nonlinear characteristics. The conventional learning curve model may also fail to predict the intermediate values accurately. Accordingly, this paper is using multiplicative calculus to estimate the parameters of the learning curve. To the best of our knowledge, there is no study using multiplicative calculus in the learning curve model. Thus, the current study is expected to fill this gap and provide comparable results to the studies that have been carried out concerning the conventional learning curve. Moreover, the results of this study are expected to be useful for managers, and cost and managerial accountants of companies using the learning effect in making decisions.

The remainder of this paper is organized as follows. Section 2 introduces multiplicative calculus and its applications to economics and finance. Section 3 presents a background for the proposed method. Section 4 contains the estimation of learning curve via multiplicative calculus. Finally, Sect. 5 presents concluding remarks and implications of the research.

2 Multiplicative calculus and its applications to economics and finance

In this section, important applications of multiplicative calculus to economics and finance are outlined in order to support the idea of integrating multiplicative calculus into the learning curve. In light of the completed applications in economics and finance, the possible impact of learning curve estimation via multiplicative based methods can be clarified.

Accordingly, it is important to start with formation of multiplicative derivative as follows: Let f with respect to x be a real positive-valued function defined in open interval (a,b). Suppose that the value changes from x to $x + h$ in the given interval.

Then, the function f changes from $f(x)$ to $f(x+h)$. Consequently, the multiplicative forward operator Δ^*, which represents the rate of change of f, can be defined as

$$\Delta^* f(x) = \frac{f(x+h)}{f(x)}$$

Multiplicative derivative, if exists, can be represented over multiplicative forward operator by

$$f^*(x) = \lim_{h \to 0} (\Delta^* f)^{1/h}$$

where $x \in \mathrm{dom}(f)$. From the formation of multiplicative forward operator Δ^*, multiplicative derivative represents rate of change. Since multiplicative derivative is scale free, it can efficiently be used for representing growth related problems. Hence, multiplicative derivative provides a natural representation of growth phenomena. For instance, we refer the study of Bashirov et al. (2008) for the following problem: *Assume that a deposit of \$(a) in a bank account grows to \$(a + d) after one day, to \$(a + m) after one month, and to \$(a + y) after one year. Then, the average daily growth factor is (a + d)/a, the average monthly growth factor is (a + m)/a, and the average annual growth factor is (a + y)/a. Also, let the instantaneous annual growth factor be p at the initial time moment. If the factor p remains constant at all-time moments during one year, then* $p = (a+y)/a = [(a+m)/a]^{12} = [(a+d)/a]^{365}$, *assuming that there are 365 days in a year and all months have equal number of days, or*

$$p = (a+y)/a = [(a+m)/a]^{(1/(1/12))} = [(a+d)/a]^{(1/(1/365))}$$

But if it is time dependent, then the above equalities become inequalities in which the expression $[(a+d)/a]^{365}$ *approximates p better than* $[(a+m)/a]^{12}$, *and also* $[(a+m)/a]^{12}$ *approximates better than (a + y)/a. Therefore, if* $f(x)$ *expresses the value of money in the bank account at the time moment x, then to find the instantaneous growth rate at the time x we have to consider multiplicative forward operator*

$$\lim_{h \to 0} (\Delta^* f)^{1/h}$$

Additionally, the geometric average of numerical data set (a_1, a_2,…, a_n)

$$\sqrt[n]{a_1 \cdot a_2 \cdot \ldots \cdot a_n}$$

is multiplicative analogue of well-known average. Chestnut (1983) also states that the geometric average is the best averaging method for stock prices because it gives excellent reasons. This shows that multiplicative (geometric) average is a natural representation of the average of stock prices. For more detailed information, we refer Bashirov et al. (2011) and Grossman (1983).

It is also important to note that elasticity can be more visible by using multiplicative derivatives. Bashirov et al. (2011, p. 428) show that the point elasticity in terms of multiplicative derivatives can be expressed as

$$f^*(x) = e^{\frac{E_{y,x}(x)}{x}}$$

where $E_{y,x}(x)$ is x arc elasticity of y. The multiplicative representation of elasticity is a very simple multiplicative differential equation. Moreover, Bashirov et al. (2011, p. 428) state that multiplicative derivative is a good tool for operating with x elasticity of y when both x and y are related to the third variable. Hence, multiplicative derivative provides a natural representation of elasticity. For more details, we refer to Bashirov et al. (2011), Cordova-Lepe (2006) and Filip and Piatecki (2014).

The function of the form

$$L(x) = ax^{-b}, \quad b > 0$$

is considered as the conventional learning function. However, this function is not linear in the classical calculus. Therefore, natural logarithm of function $L(x)$ should be considered in the estimation process. To illustrate, Peles (1991, p. 356) transformed this function into a logarithmic form in order to estimate parameters of learning curves. It is important to emphasize that the conventional learning function $L(x)$ is linear in the multiplicative calculus so it does not need logarithmic transformation. Hence, the methods based on multiplicative calculus could be appropriate for analyzing learning curve which is exponential function.

If we also consider the classical least square method

$$S(x) = \sum_{i=0}^{n} \left(y_i - ax_i^{-b}\right),$$

for the estimation of a and b in $L(x)$, natural logarithm transformation is necessary. Therefore, Peles (1991), for example, followed this transformation in his study. However, this transformation increases time consumption and complexity. On the other hand, the multiplicative least square method, which was introduced by Özyapıcı and Bilgehan (2015),

$$M(x) = \prod_{i=0}^{n} \left(\frac{y_i}{ax_i^{-b}}\right)^{\ln\left(\frac{y_i}{ax_i^{-b}}\right)}$$

can directly be applicable for the estimation of a and b without logarithmic transformation. For more details about multiplicative numerical methods, we refer to Özyapici et al. (2014) and Riza et al. (2009).

3 Exponential approximation: a background for the proposed method

3.1 Multiplicative Lagrange interpolation

If we consider the two data points $(x_0, f(x_0))$ and $(x_1, f(x_1))$, then the exponential function passing through these two points can be defined as follows:

$$E_1(x) = [f(x_0)]^{\frac{x-x_1}{x_0-x_1}} \cdot [f(x_1)]^{\frac{x-x_0}{x_1-x_0}}$$

where it satisfies $E_1(x_0) = f(x_0)$ when x = x_0 and $E_1(x_1) = f(x_1)$ when $x = x_1$.

For the more general situation, when $(n + 1)$ data points are given, say $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_n, f(x_n))$, we can define the multiplicative Lagrange interpolation to be used for computing learning curve values as follows:

$$E_n(x) = [f(x_0)]^{L_{n,0}(x)} \cdot [f(x_1)]^{L_{n,1}(x)} \cdots [f(x_n)]^{L_{n,n}(x)} = \prod_{k=0}^{n} [f(x_k)]^{L_{n,k}(x)} \tag{1}$$

where

$$L_{n,k}(x) = \prod_{\substack{i=0 \\ i \neq k}}^{n} \frac{(x - x_i)}{(x_k - x_i)}$$

for each $k = 0,1,\ldots,n$. The error of approximation $E_n(x)$ was given by (Misirli and Özyapici 2009) as

$$E(x) = \left[f^{*(n+1)}(\xi)\right]^{\prod_{i=0}^{n} \frac{(x_i - x_k)}{(n+1)!}}$$

where f is $(n + 1)$th*-differentiable on [a,b] and $a < \xi < b$.

Since interpolation techniques are widely used in science, engineering, finance and economics, multiplicative Lagrange interpolation provides a wide range of applications with very efficient estimations in many problems.

3.2 Learning curve and multiplicative Lagrange interpolation

The classical derivative of a function f represents the slope of the tangent line to the graph of f at points $(x_0, f(x_0))$, i.e. $f(x) \sim f(x_0) + f'(x_0)(x - x_0)$.

Analogously, multiplicative derivative of a function f represents multiplicative slope of multiplicative uniform function

$$f_m(x) = ax^b.$$

The function $f_m(x)$ is also assumed as conventional learning function. Hence, it is important to note that graphical meaning of multiplicative derivative also matches with the learning function. Additionally, $f_m(x)$ can be approximated around x_0 in the multiplicative calculus as

$$f_m(x) \sim f_m(x_0)\left(f_m^*(x_0)\right)^{x-x_0}.$$

As an application of multiplicative calculus to the learning curve, when the study conducted by Peles (1991) is considered, the exponential function $f(x) = a_1 e^{-b_1 x}, b_1 > 0$ which is the best fit to the learning curve estimation

$$L(x) = 0.6768x^{-0.322} \text{ around } x_0 = 1$$

can be obtained as

$$L(x) \approx E(x) = 0.678e^{-0.322(x-1)} = 0.9355e^{-0.322}.$$

On the other hand, corresponding classical approximation of learning curve analyzed by Peles (1991) can be given as

$$L(x) \approx C(x) = 0.678 - 0.2183(x-1) = 0.8963 - 0.2183x.$$

Accordingly, it is not difficult to verify that exponential function $E(x)$ is better than classical linear function $C(x)$ for the estimation of the learning curve. Nevertheless, this should not be surprise since learning curve estimation can perfectly be fitted by decreasing exponential function. Hence, this proves that multiplicative Lagrange interpolation method (1) can be effectively used for interpolating learning data. The efficiency of method (1) can also be observed in the next section.

4 Estimation of learning curve via multiplicative calculus

As stated by McDonald (1987), cost data may not be available for each unit of product produced for several reasons such as two or more products are produced simultaneously or the cost contract is related to the whole production rather than a particular product. The learning curve estimation is therefore needed. To better understand this issue, consider a company in which the production takes place after an order has been received. In order to estimate costs by using the learning curve phenomenon, all values should be recorded. However, because of the irregular characteristics of the orders, some of intermediate values of cumulative production cannot be recorded. For instance, assume that the number of order on the first day is 20 whereas it is 50 on the second day. If the company attempts to calculate the labour cost to be incurred for producing the order of third day which is 30 units, the company should estimate the learning effect of this specific intermediate value based on the observed data of the 20th and 50th units. However, the conventional learning curve estimations do not produce accurate results since the learning curve equation is a decreasing exponential function. In this respect, multiplicative based method is likely to produce more accurate results since it provides a better fit than the conventional nonlinear methods.

In order to clarify the above discussions, two examples, which demonstrate the superiority of the multiplicative Lagrange interpolation for learning curve estimation, will be considered. The first example was taken from the article written by Peles (1991, p. 352–355), while the second example was obtained from one of the most popular management and cost accounting books written by Drury (2012, p. 619–622).

Example 1 Peles (1991, p. 349–359) criticized the learning curve since it ignores the effect of time on costs. Accordingly, the researcher improved the learning curve in the accounting literature by adding the rate effect to the conventional learning curve

equation. Nevertheless, the alternative method that Peles (1991) proposed was still unable to calculate intermediate values accurately. For the purpose of this study, the learning curve equation given in Peles's paper (p. 324) is used to compare with our proposed learning curve equation which is combined by multiplicative Lagrange.

Table 1 shows the learning curve values that are representing the learning function used by Peles (1991). Peles (1991) assumed a company has to produce new products. Accordingly, Table 1 shows both the first 10 units of products to be produced and their costs given by Peles (1991). In order to test multiplicative based learning curve estimation, we removed data from production levels randomly. It is important to emphasize that which data will be removed from the list is not important. This is because the learning curve data present exponential behaviour on graphics. In this regard, multiplicative based methods can work well since exponential functions are linear in the multiplicative calculus.

We first assumed costs of producing 2 units, 4 units, and 7 units, respectively, are not known. Then, we assumed both 2 and 7 units simultaneously are not known in this production process. Finally, we removed three units that are unit 3, 5, and 6 at the same time, and made the estimations. We first used conventional learning curve equation to estimate costs of these units that are missing. Then, we used classical Lagrange interpolation and multiplicative Lagrange interpolation so as to compare the results. Table 2, therefore, presents a comparison of the accuracy of estimates of conventional learning curve, classical Lagrange interpolation, and multiplicative Lagrange interpolation methods.

As can be seen in Table 2, the proposed learning curve method, which was combined by multiplicative Lagrange, was tested when some of the units given in Table 1 were missing. Even though Peles's approach has improved the learning curve equation, the above results demonstrate that our proposed method gives better

Table 1 Periodic cost data given by Peles's article

Unit	1	2	3	4	5	6	7	8	9	10
Periodic cost	1.000	0.600	0.506	0.454	0.418	0.392	0.371	0.354	0.341	0.328

Table 2 Cost estimations of missing units

Missing units	Actual value used by Peles	Conventional method used by Peles	Classical Lagrange interpolation	Multiplicative Lagrange interpolation
Unit 2	0.600	0.542	0.613	0.602
Unit 4	0.454	0.434	0.455	0.454
Unit 7	0.371	0.362	0.370	0.371
(Unit 2	0.600	0.542	0.622	0.610
Unit 7)	0.371	0.362	0.372	0.371
(Unit 3	0.506	0.475	0.485	0.495
Unit 5	0.418	0.403	0.428	0.423
Unit 6)	0.392	0.380	0.398	0.394

results. Table 2 also shows that, according to our proposed method, the intermediate values of 2nd, 4th, and 7th units, which are assumed to be missing values, are closer to reality when compared with the values found by both Peles' learning curve function and classical Lagrange method. For example, when unit 2 is missing, our proposed method estimates periodic cost as 0.602 which is closer to actual value (0.600) than the values (0.542 and 0.613) estimated by conventional learning curve and classical Lagrange interpolation methods, respectively. This is also the case when two or three units are missing within the data set.

Example 2 Drury (2012) presents an application of the 80 % learning curve which means that 20 per cent productivity improvement occurs each time production is doubled. According to the example given by Drury (2012, p. 619–622), a sequence of six orders where the cumulative number of units is doubled for each order was given. The author emphasized that other quantities such as 10, 20, or 30 units which are not known can be calculated mathematically by using the learning curve equation as follows:

$$Y_x = 2000x^{-0.322} \tag{2}$$

In this respect, Drury (2012) uses the function (2) to calculate the time needed to produce each unit of cumulative production. Drury (2012) considered 80 per cent learning curve and calculated the values representing the learning curve function. Thus, the cumulative hours for the cumulative production of 1, 2, 4, and 8 units of products are found to be 2000, 1600, 1280, and 1024 h, respectively. Drury (2012) stated that since the cumulative number of units is doubled, no information is available for other quantities such as, 3, 5, 6, and 7 units and the learning function (2) can be used to estimate these values that are missing.

Table 3 shows both the cumulative hours for the cumulative production of 1, 2, 4, and 8 units and the estimations of intermediate values that are the time needed to produce 3, 5, 6, and 7 units in the production process, respectively. We first used conventional learning function considered and emphasized by Drury (2012) to estimate values of intermediate units that are missing. Then, we used classical Lagrange interpolation and multiplicative Lagrange interpolation in order to compare the results.

Table 3 Intermediate learning curve estimation using the data obtained by Drury's article

Number of units	Conventional method used by Drury	Classical Lagrange interpolation	Multiplicative Lagrange interpolation
1	2000	2000	2000
2	1600	1600	1600
3	1404	1378	1389
4	1280	1280	1280
5	1191	1250	1220
6	1123	1234	1175
7	1068	1177	1117
8	1024	1024	1024

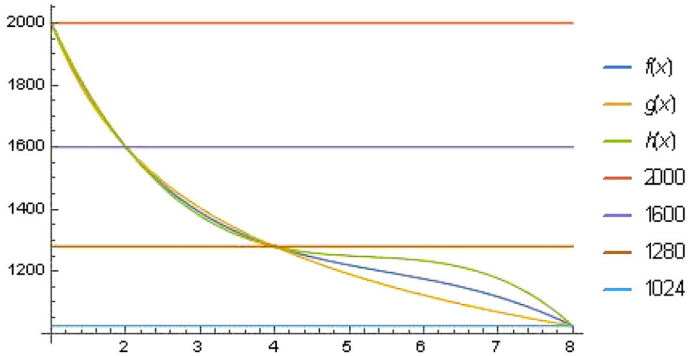

Fig. 1 Graphical representation of learning curve estimation methods for Drury's data, *f(x)* = Drury's learning curve, *g(x)* = Learning curve produced by multiplicative Lagrange interpolation, *h(x)* = Learning curve produced by classical Lagrange interpolation

According to the estimations of intermediate values that we obtain, the learning curve equation based on multiplicative Lagrange produces closer results to actual values than that of conventional equation as well as classical Lagrange. This is due to the fact that exponential functions are linear in multiplicative calculus and hence they produce better results in estimations via multiplicative calculus. Thus, the time needed to produce intermediate units will be calculated more accurately. This leads cost and managerial accountants to estimate both cost of production and unused labour capacity more precisely. In this context, the selling price, which is based on the cost data calculated by using the conventional learning curve method, may also need to be revised based on these results. The employee-related capacity decisions such as the number of employees to be dismissed or directed to other areas may also need to be revised in accordance with the results found by multiplicative based learning curve method.

Figure 1 portrays the graphical representation of conventional method used by Drury, multiplicative Lagrange interpolation and classical Lagrange interpolation that are represented by $f(x)$, $g(x)$ and $h(x)$, respectively.

From the graphical representations of the learning data given in Fig. 1, an exponential representation can be made. This is the general situation for many learning curve estimations. Since multiplicative interpolation uses exponential functions, it provides a better fit than the conventional methods. Comparing the results found by Peles (1991) and Drury (2012) to the ones produced by this study will help an audience better understand the efficiency of multiplicative Lagrange interpolation method. All of these also lead to the conclusion that multiplicative Lagrange based learning curve equation is able to yield more accurate results than the ones produced by conventional learning curve equation.

5 Conclusion

The purpose of this study was to develop a way to estimate the learning curve using multiplicative calculus in order to see whether the multiplicative based learning curve better estimates the intermediate values of cumulative production than the

ones obtained by the conventional learning curve. The findings derived from this study demonstrate that the new approach to the learning curve is able to calculate the intermediate values better than the conventional learning curve approach. Therefore, this study makes a contribution to the literature by highlighting the efficiency of multiplicative Lagrange interpolation in the estimation of intermediate values.

The learning curve is a management tool used in cost estimation and planning. Because of three important reasons, we proposed a learning curve equation created with multiplicative Lagrange for managers, and cost and managerial accountants. These three reasons are as follows: (1) the literature review reveals that the learning rate is generally considered constant. However, the learning rate may vary due to physical, psychological, and environmental factors in a real life application. Thus, the conventional learning curve which is based on a constant learning rate may fail to satisfy the flexibility requirement of the real-life applications. (2) All intermediate values are not available in real applications since the orders or the numbers of customers are likely to be irregular. Thus, an interpolation method which is mathematically proven is needed to estimate the intermediate values efficiently. (3) The learning curve is fitted with an exponentially decreasing function. In this regard, the multiplicative calculus is able to better fit to the learning curve.

The results of the present study are expected to provide important managerial implications to decision makers. First, companies especially having high employee turnover may use multiplicative Lagrange based learning curve equation to accurately estimate intermediate values for new employees. This will in turn enable cost and managerial accountants to better calculate direct labour hours/costs and product costs. Calculation of the product costs more accurately is expected to improve decision-making concerning such issues as cost-volume-profit analysis, pricing, and standard costing. The calculation of direct labour hours/costs more accurately is also expected to assist business managers and cost and managerial accountants in determining unused capacity, making labour planning, and budgeting. Second, the results are expected to help managers better estimate costs and recognize revenues for potential contracts. Third, since labour costs need to be considered together with the learning effect, the results are also expected to help cost and managerial accountants of companies implementing Time Driven Activity Based Costing System calculate labour costs more accurately. Fourth, the results of this study also suggest that researchers interested in economics and finance are able to use multiplicative Lagrange interpolation for better estimations of data represented by nonlinear functions. Accordingly, researchers using learning curve or exponential models in the fields of economics, finance, and business may utilize multiplicative Lagrange interpolation for improving the results of future studies.

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